

VM14K — Manual Verification of Contradiction Groups

Method. 25 contradiction groups were reviewed by hand against `data-processed-shuffled0.jsonl`. Groups were classified by *failure type* rather than treated as a single “contradiction” category, because inspection showed the flagged set contains at least six distinct defects.

Summary of the 25 reviewed groups

Category	Groups	Count
Genuine answer-key contradictions	3, 15, 36, 37, 46, 153, 154, 176, 197	9
Different option sets (false positives)	40, 108, 110, 165, 191	5
Multi-answer item exploded into rows	9	1
Image-dependent, unanswerable	55, 65	2
Encoding / formatting / data-entry corruption	49, 59, 141, 173, 183, 184, 185, 189	8
(Overlapping: true/false truncation)	43, 180	2

Findings by category

1 Genuine answer-key contradictions 9 of 25

Same question, same option set, different marked answer. At least one of each pair is wrong.

Group	Notes
3	Direct contradiction, A vs D
15	4 duplicates: 3 true dupes + 1 with a different answer
36	Same Q and options; conflicting answers AND conflicting difficulty labels
37	Near-duplicate of Group 36; conflicting answers
46	Many duplicates, identical throughout; one conflicting answer
153	Conflicting answers, same Q
154	Same as 153
176	Conflicting answers
197	Conflicting answers

These are the defensible core. Only this category supports the claim “the answer key contradicts itself.”

2 Option sets differ, so answers legitimately differ 5 of 25

NOT contradictions. Flagged only because the grouping key used question text alone.

Group	Notes
40	Same question, entirely different option sets. Also: one option is literally "D....." — a truncation artifact. Unclear whether Lao màng bụng and Viêm màng bụng do lao are meant to be the same answer.
108	Duplicates sit adjacent in the file; same question, different options (“choose the wrong statement” variants)
110	Same pattern as 108. Answers are consistent with their respective option sets
165	Same Q, different option ranges — plus contains a “Cả A và B” option (see Category 6)
191	Same Q, different options — likely also explains the difficulty difference

These are **false positives** in the contradiction count and must be excluded from any error-rate calculation.

3 Multi-answer question with different subsets of options 1 of 25

Group 9 — “Đặc điểm tổn thương vi thể trong viêm cầu thận cấp?” The copies carry *different option subsets*. Merging all options across copies recovers what looks like an 8-option multi-select item:

- Cầu thận teo nhỏ, nhiều tế bào
- Cầu thận teo nhỏ, ít tế bào
- Cầu thận nở to, nhiều tế bào
- Cầu thận nở to, ít tế bào
- Khoảng Bowman hẹp, có tế bào viêm, hồng cầu
- Khoảng Bowman rộng, có tế bào viêm, hồng cầu
- Lòng ống thận rộng, có hồng cầu, tế bào viêm, protein
- Lòng ống thận hẹp, không có hồng cầu, tế bào viêm, protein

ChatGPT 5.5 asked independently identified **three** of these as correct.

4 Image-dependent questions with no image 2 of 25

Unanswerable by any text model. Every model is scored wrong regardless of competence.

Group	Notes
55	Question is literally “đây là gì” (“what is this?”). No image exists in the dataset. Answers across copies are wildly different.
65	Question indicates an image is required; different options across copies

The paper never mentions image-based questions anywhere.

5 Encoding, formatting, and data-entry corruption 8 of 25

Group	Defect
184	Option text ran together across a line break . Option A reads “Cạp vào niêm mạc ruột vật chủ và hút máu B. Thuỷ phân mô vật” — swallowing the “B.” marker and half of option B. Option B is the orphaned tail “chủ và hút máu”.
185	Question has exactly one option : [“ <i>Ascaris lumbricoides</i> và <i>Ancylostoma duodenale</i> ”]. Unanswerable as an MCQ. Its duplicate is a normal 4-option question.
59	Typos in the answer text across copies: Cáo Nitrosamin vs Các Nitrosamnín
183	Unicode inconsistency: B ₁₂ (subscript) vs B12 (ASCII). Not a contradiction — a normalisation failure.
49	Trailing-space/period formatting in options (“Biệt hoá cao .”)
173	Answers differ only by wording, not by meaning
141	True dupes, same answer; differ only in spacing/punctuation — but carry different difficulty labels

Group	Defect
189	Questions differ only by a final “:” vs “?” . Same options. Different difficulty AND different marked answer.

6 Position-referential options (shuffle-breaking)

Group 165 contains an option reading “Cả A và B” (“Both A and B”).

Concern: If an option’s text names other options *by letter*, then shuffling the option order **invalidates the question** — the letters now point at different content, and the correct answer can silently change. The dataset ships three files with shuffled option orders, so two of them contain semantically broken versions of every such item.

Confirmed programmatically: **at least 80 questions** in the released data contain a position-referential option.

7 True/false items with mangled option text 2 of 25

Group 43 — 4 duplicates of a 2-option question. Options are “Đ” / “Đúng” and “S” / “Sai” (false). Also carries inconsistent difficulty labels (one Easy, three Medium).

Group 180 — same Đúng/Sai truncation problem.

These are true/false questions with a **50% random baseline**, not 25%. The dataset contains **1,240 two-option questions**, all scored in a single aggregate accuracy figure alongside 4-option items.

Cross-cutting: difficulty labels are unstable on identical items

Observed in Groups **36, 43, 60, 61, 62, 141, 189** — the same question, with the same options, receives *different* difficulty labels across copies.

Group	Observation
60	One copy Challenging, one Medium
61	Same pattern
62	One copy Challenging , one Easy — despite identical question and option set

This is a measurable reliability failure in the LLM difficulty labeller. It matters disproportionately because the **Challenging tier has only 1,193 items and the Hard tier only 90**. Instability at the top tiers undermines the paper’s four-tier framework.

What still needs checking

- The true count of image-dependent questions. Only 2 appeared in a 25-group sample; the full-dataset count is unknown.
- Whether Group 9’s multi-select-explosion pattern is widespread.

- The actual correct answers for the conflicting-answer questions.